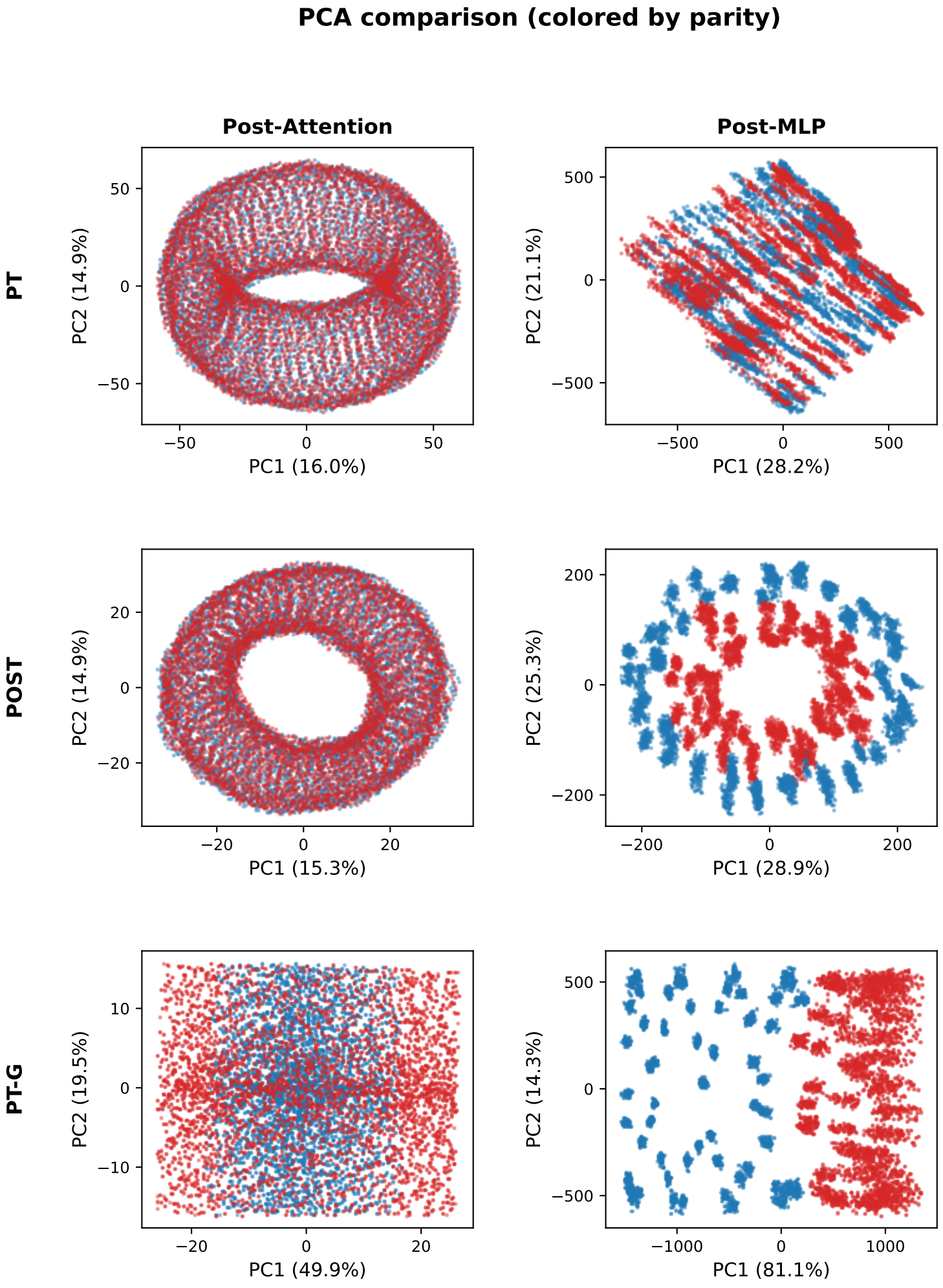
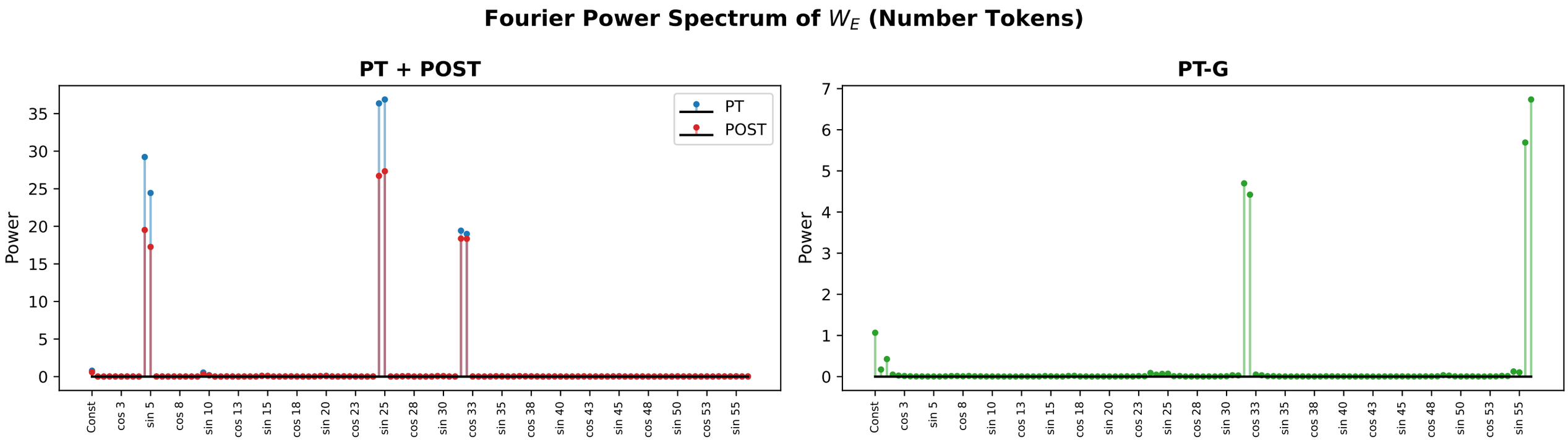
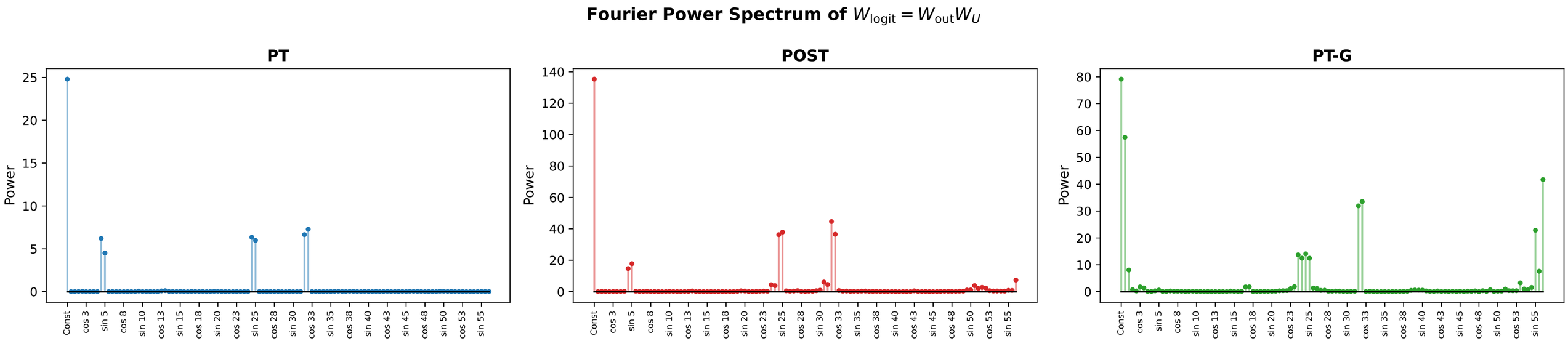
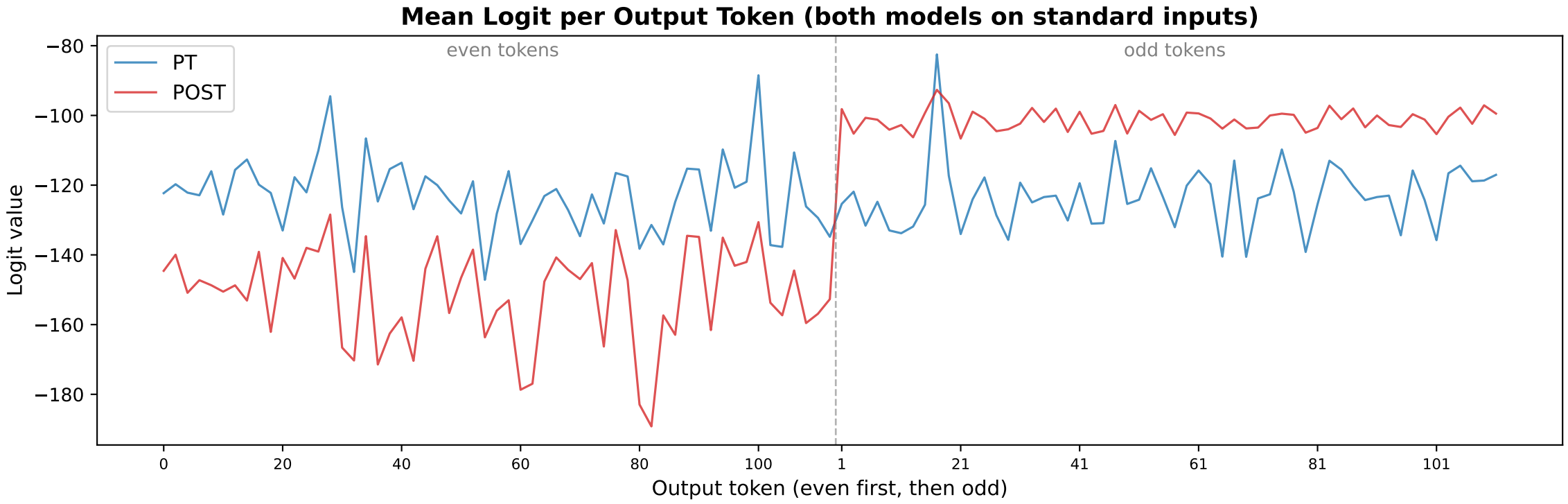
Test Performance (Parity-Gated Evaluation)

	PT	POST	PT-G
Test loss	1.84e-07	2.08e-07	6.82e-07
Test acc	1.0000	1.0000	1.0000
Even acc	1.0000	1.0000	1.0000
Odd acc	0.0000	1.0000	1.0000


PCA comparison (colored by parity)
Post-Attention
Post-MLP
PT
POST
PT-G
PC1 (16.0%)
PC2 (14.9%)
PC1 (28.2%)
PC2 (21.1%)
PC1 (15.3%)
PC2 (14.9%)
PC1 (28.9%)
PC2 (25.3%)
PC1 (49.9%)
PC2 (19.5%)
PC1 (81.1%)
PC2 (14.3%)
even
odd


Fourier Power Spectrum of W_E (Number Tokens)
PT + POST
PT
POST
Power
0
5
10
15
20
25
30
35
Const
cos 3
sin 5
cos 8
sin 10
cos 13
sin 15
cos 18
sin 20
cos 23
sin 25
cos 28
sin 30
cos 33
sin 35
cos 38
sin 40
cos 43
sin 45
cos 48
sin 50
cos 53
sin 55
PT-G
Power
0
1
2
3
4
5
6
7
Const
cos 3
sin 5
cos 8
sin 10
cos 13
sin 15
cos 18
sin 20
cos 23
sin 25
cos 28
sin 30
cos 33
sin 35
cos 38
sin 40
cos 43
sin 45
cos 48
sin 50
cos 53
sin 55


Fourier Power Spectrum of $W_{\text{logit}} = W_{\text{out}} W_U$
PT
POST
PT-G
Power
Const
cos 3
sin 5
cos 8
sin 10
cos 13
sin 15
cos 18
sin 20
cos 23
sin 25
cos 28
sin 30
cos 33
sin 35
cos 38
sin 40
cos 43
sin 45
cos 48
sin 50
cos 53
sin 55


Mean Logit per Output Token (both models on standard inputs)
PT
POST
even tokens
odd tokens
Logit value
−80
−100
−120
−140
−160
−180
0
20
40
60
80
100
1
21
41
61
81
101
Output token (even first, then odd)